mbLP

Deposit BTC. Hold one token. Earn MEZO as BTC/MUSD liquidity.

One transaction opens a MUSD trove, borrows, adds liquidity and stakes it in the gauge. The vault runs the trove for you. Exposure stays about 1x BTC. No owner, no admin key.

Mezo Buildathon · Wave 1 · mblp.surge.sh · github.com/GuTS805/Akindo

THE PROBLEM

Mezo holds a lot of BTC, but BTC/MUSD liquidity is thin

806 BTC

locked in veBTC

~1 BTC

of BTC/MUSD liquidity across all pools

1–2%

cost to buy BTC with MUSD at $1k–$10k, against the oracle

Mezo mainnet, read on-chain 19–20 Oct 2026

WHY IT STAYS THIN

The rewards are there. The work is the problem.

To provide BTC/MUSD today you

1. open a MUSD trove with BTC
2. borrow MUSD
3. add MUSD and BTC to the pool
4. stake the LP in the gauge
5. watch the collateral ratio, forever

42,048

MEZO a day already paid to the volatile BTC/MUSD gauge

Gauge emissions read from Mezo mainnet, Oct 2026

MBLP

One BTC deposit does all five steps

Deposit

Plain BTC in, mbLP shares out, priced from the Mezo oracle

Trove

~75% of BTC as collateral at 300% CR; the vault borrows MUSD

Pool

MUSD plus the other ~25% of BTC into BTC/MUSD on Mezo Pools

Gauge

LP staked; MEZO credited per share, claim any time

The vault keeps the ratio in its 250–350% band without swaps. Exposure stays about 1x BTC.

Debt is held at NAV / (target CR + 1): at target, collateral = 3× debt and the LP holds the debt’s worth of MUSD and of BTC

REWARDS

MEZO streams to every share, every second

The gauge pays the vault. **harvest()** takes a 10% fee and credits the rest per share, before any share moves, so claims stay exact across transfers.

On testnet the default vault (v3) stakes in Mezo’s live BTC/MUSD gauge, which Mezo’s Voter pays each epoch (760,811 MEZO on 22 Oct), so it earns real emissions exactly as on mainnet.

652

MEZO a day on 1 BTC from Mezo’s live testnet gauge (fork test)

Claimable MEZO ticks up live in the app; anyone, or the keeper, can harvest

v3 vault 0x75532765bfBa32E8DC796292937F9c571a2dBFEC · gauge 0x72D88B0795373437E86792DEd8362D66dC43Cd2c

MEZO EARN

The vault’s fees buy votes for its own pool

More BTC
deposited, more staked LP

More fees
10% of the MEZO harvested

Incentives
posted on the BTC/MUSD gauge

More votes
from veBTC, so more emissions

184

MEZO posted on mainnet’s real gauge incentive contract from one day of fees, on a fork. Live on testnet since 22 Oct

An **IncentiveRouter** as the fee recipient splits each fee between gauge incentives and the treasury. No owner; anyone can push. This is the vote-escrow side of Mezo Earn, used the way the whitepaper intends.

contracts/src/IncentiveRouter.sol · tested on the live BribeVotingReward 0x94A9…E305 in MainnetFork.t.sol

DESIGN

Built for a thin market

Rebalance never swaps

Borrows, repays and moves BTC between trove and pool to hold the target ratio.

Oracle band

Every pool action checks the pool price against Mezo’s oracle and waits if it drifts.

Fair LP value

Shares price LP at $2 \cdot \sqrt{(k \cdot \text{price})}$, which trading the pool cannot move.

Exits leave the pool alone

A leaver’s MUSD gap settles inside the vault at the oracle price + 0.5%.

Nobody gets stuck

An exit that would break MUSD’s minimum debt unwinds the position instead.

No owner

Fixed parameters, no upgrades; liquidation and redemption recover on their own.

Full mechanics and the Slither review: contracts/README.md and contracts/SECURITY.md

MEZO INTEGRATION

Built on MUSD, Mezo Pools, MEZO and Passport

Mezo piece	What mbLP does with it
MUSD	One trove via BorrowerOperations, TroveManager, HintHelpers, SortedTroves
PriceFeed	Prices shares, bands every pool action, settles exits at the oracle price
Mezo Pools	Router, volatile BTC/MUSD pool and its gauge
MEZO	The reward the vault harvests and pays out per share
BTC precompile	Native BTC is the only asset in and out, and the pool’s ERC-20 side
Mezo Passport	Bitcoin wallets (Unisat, OKX, Xverse) act through a Mezo smart account
MUSD events	After liquidation: plain LP, then reopens. After redemption: surplus reclaimed

Testnet and mainnet addresses from mezo-org/musd and mezo-org/tigris, each checked on chain

WHY A VAULT

Routing into LP is one step. mbLP runs the position.

You need to	Route or zap tools	mbLP
Bring	MUSD, or a trove you manage	Plain BTC only
Watch the collateral ratio	Yes, yourself	No: the vault rebalances, anyone can trigger it
Earn gauge MEZO	Only if you stake the LP yourself	Yes: staked, harvested, split per share
Handle liquidation or redemption	Yourself	The vault recovers and reopens
Exit	Unwind each step	One withdraw, settled inside the vault

Exposure stays about 1x BTC either way; the difference is who does the work

PROOF, LIVE

A fresh wallet ran the full cycle on testnet

Deposit

0.005 BTC: borrow, add LP, stake and rebalance in one transaction

Claim

MEZO paid to the wallet ten minutes later

Withdraw

Half the shares, 0.0025 BTC back

A real Unisat wallet then ran the same cycle through Mezo Passport, after the app worked around Passport’s dead testnet relayer host (Mezo confirmed the fix). The first run also caught Mezo’s gas estimate falling short on exits, so every vault transaction now gets headroom.

Every transaction hash: docs/evidence.md in the repository

MAINNET, ON A FORK

Against Mezo mainnet's real MUSD, pool and MEZO gauge

3,309

MEZO earned in one day on a 0.2 BTC deposit

0.19995

BTC back on the full exit

The mainnet configuration already exists; this test runs in CI on every push.

contracts/test/MainnetFork.t.sol · MezoMainnet in contracts/script/Addresses.sol

TESTED HARD

Every claim has a test behind it

78

unit, fuzz, manipulation, recovery, gauge, router and edge-case tests

65,536

random invariant actions, then every holder exits

97%

line coverage, 100% of functions

41/41

Slither results reviewed, none needed a change

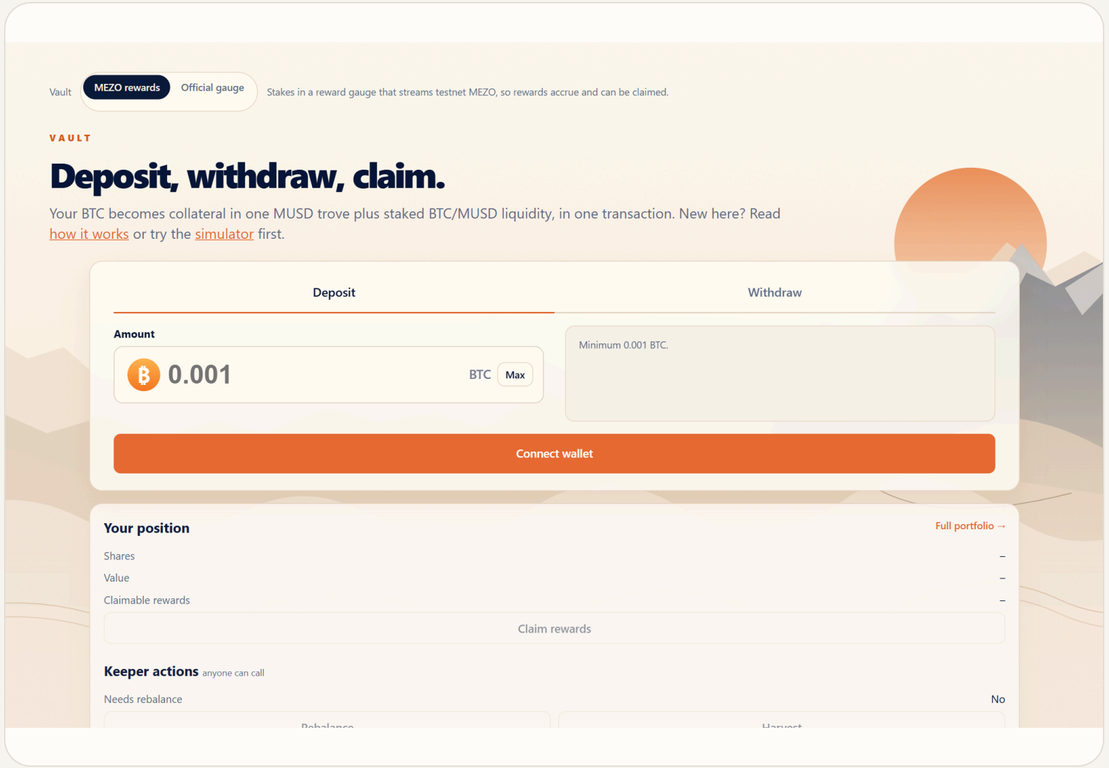
Fork tests on testnet and mainnet run in CI on every push. The tests found 5 real bugs before deployment, and a review pass led to v2, now live: deposits pay their own borrowing fee, a deposit cap, and a failed deployment never blocks a deposit.

contracts/test · contracts/SECURITY.md · not audited

THE APP

Six pages, all read from chain

- **Vault**: deposit, withdraw with the exact payout simulated, claim
- **Portfolio**: position, result vs deposits, MEZO ticking up live
- **Analytics**: share price and rewards history, live mainnet market
- **Simulator**: mbLP vs holding BTC
- **Home** and **Docs**
- EVM wallets or Bitcoin wallets via Mezo Passport



YIELD

What the same strategy earns on mainnet today

~55%

LP APR in MEZO on the volatile BTC/MUSD gauge

~24%

for depositors: ~half of value is LP, minus the 10% fee, before 1% MUSD interest

1x

BTC exposure kept: BTC plus yield, not leverage

The mainnet fork agrees: 3,309 MEZO a day on 0.2 BTC is about 22% a year at today’s MEZO price. The rate falls as liquidity joins the gauge, which is the point.

Read live in the app from the mainnet gauge and the MEZO/BTC pool; 20 Oct 2026, MEZO ≈ $0.0029

HONEST LIMITS

What a depositor should know

- Rewards are MEZO, which has little on-chain liquidity; the vault pays them out and never sells them
- Impermanent loss on the ~25% of value in the pool
- While Mezo’s oracle is stale or MUSD is in recovery mode, some trove changes and exits wait
- The exit that closes the trove waits for arbitrage rather than sell BTC below the oracle band
- In some states after a redemption or liquidation, a deposit’s deployment fails its CR check and waits as idle, withdrawable BTC (safe; fix planned)
- Unaudited hackathon code

Each limit is documented, with its reason, in the repository README; the last-exit case has its own test

Wave 2, then the road to mainnet

1. Concentrated BTC/MUSD pool, the gauge’s larger share
2. Optional MEZO → veMEZO compounding
3. Then freeze, audit and a capped mainnet launch on Mezo’s gauge (config and fork test ready)

Try it now

mblp.surge.sh

github.com/GuTS805/Akindo

Evidence: docs/evidence.md

Deposit BTC. Deepen Mezo.